🎢 005 Week 2 — English Worksheet (Intermediate)

Topic: One Function, Many Booths — Arguments & Parameters · **Course:** Theme Park Creator · **Time:** about 45 minutes

This week you build a park **booth** with a **function**. You give it **parameters** — the materials and size it uses.

Change the **arguments** you **call** it with, and the **same** function builds a wood booth, a stone booth, a big one or a small one.

These are the blocks you use this week:

```
blocks.fill(OAK_PLANKS, pos(0, 0, 0), pos(4, 0, 4), FillOperation.HOLLOW)
blocks.fill(STONE, pos(0, 0, 0), pos(4, 0, 4), FillOperation.REPLACE)
blocks.fill(AIR, pos(2, 0, 0), pos(2, 1, 0), FillOperation.REPLACE)
```

> `blocks.fill(material, corner1, corner2, ...)` fills a box between two corners. `pos(x, y, z)` is a spot: `x` = left/right, `y` = up, `z` = forward/back. `FillOperation.HOLLOW` = only the outer **walls**. `FillOperation.REPLACE` = the **whole solid** box. `AIR` = empty space. Fill a slot with `AIR` to **carve a doorway** out of a wall.

1 · Meet Arguments and Parameters 🎁

A **parameter** is an empty named box — you make it when you *define* the function. An **argument** is the real value you drop in — you send it when you *call* the function. They line up **by position**, left to right.

```
MAKE IT  →  the names in ( ) are PARAMETERS (empty named boxes):

    def build_booth( wall , floor , size ):

CALL IT  →  the values in ( ) are ARGUMENTS (real values sent in):

    build_booth( OAK_PLANKS , STONE , 5 )

Position matches — 1st→1st, 2nd→2nd, 3rd→3rd:

    wall   ⟵  OAK_PLANKS     (1st box gets the 1st value)
    floor  ⟵  STONE          (2nd box gets the 2nd value)
    size   ⟵  5              (3rd box gets the 3rd value)
```

Why does one function with parameters let you build many different booths?

2 · Predict 🔮

Read each function. Write what you think happens.

```
def build_booth(wall, floor, size):
    blocks.fill(floor, pos(0, 0, 0), pos(size, 0, size), FillOperation.REPLACE)
    blocks.fill(wall, pos(0, 0, 0), pos(size, 3, size), FillOperation.HOLLOW)

build_booth(OAK_PLANKS, STONE, 5)
```

What material is the floor? What material are the walls? How wide is the booth?

```
def build_booth(wall, floor, size):
    blocks.fill(floor, pos(0, 0, 0), pos(size, 0, size), FillOperation.REPLACE)
    blocks.fill(wall, pos(0, 0, 0), pos(size, 3, size), FillOperation.HOLLOW)

build_booth(GLASS, STONE, 3)
build_booth(BRICKS, OAK_PLANKS, 8)
```

The same function is called twice with different arguments. Are the two booths the same? Which one is bigger?

3 · Spot the Bug 🐛

Each block is broken. Read what it should do, rewrite it so it works, then explain why the original was wrong.

Bug A — This should build one booth. But when you run it, **nothing happens at all**.

```
def build_booth(wall, floor, size):
    blocks.fill(floor, pos(0, 0, 0), pos(size, 0, size), FillOperation.REPLACE)
    blocks.fill(wall, pos(0, 0, 0), pos(size, 3, size), FillOperation.HOLLOW)
```

Hint: writing `def` only **teaches** the function. Something is missing after it.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The booth should be as wide as the number you send in. But `build_booth(OAK_PLANKS, STONE, 9)` and `build_booth(OAK_PLANKS, STONE, 2)` build the **same size** every time.

```
def build_booth(wall, floor, size):
    blocks.fill(floor, pos(0, 0, 0), pos(4, 0, 4), FillOperation.REPLACE)
    blocks.fill(wall, pos(0, 0, 0), pos(4, 3, 4), FillOperation.HOLLOW)

build_booth(OAK_PLANKS, STONE, 9)
```

Hint: look at the numbers inside. The function never uses `size`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This should build a **stone floor** with **glass walls**. But it comes out with glass on the floor and stone walls — the wrong way around.

```
def build_booth(wall, floor, size):
    blocks.fill(floor, pos(0, 0, 0), pos(size, 0, size), FillOperation.REPLACE)
    blocks.fill(wall, pos(0, 0, 0), pos(size, 3, size), FillOperation.HOLLOW)

build_booth(STONE, GLASS, 5)
```

Hint: the arguments line up with the parameters **by position**. Which value landed in which box?

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Build the Booth 🏪

In your homework world, write **one** function called `build_booth` that builds a booth a visitor can walk into. Your function must have these **three parameters**:

```
def build_booth(wall, floor, size):
```

Your booth must include:

- a **floor** made from the `floor` material, `size` wide
- **walls** made from the `wall` material (use `FillOperation.HOLLOW` so the middle stays empty)
- a **doorway** carved out of one wall with `AIR` so a visitor can walk in

Then **call your function twice** with different materials and sizes, for example:

```
build_booth(OAK_PLANKS, STONE, 5)
build_booth(BRICKS, OAK_PLANKS, 8)
```

Write your finished `build_booth` function and the two lines that call it:

5 · Show Your Work 📸 🎥

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

> Today I built _____.
>
> I built it using this code: _____.
>
> In this code I used _____.
>
> The hardest part was _____.
>
> That part was hard because _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.